

# INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

## THE SITE

Location Amazon, CA -- station KSJC, America/Los\_Angeles  
Committed pair OSM 1065780801 -> 209117217  
Amazon / Evocative Santa Clara SJC11  
Facade gap 281.3 m between the two halls  
Weather record 43,747 real hourly records from KSJC  
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.2872 C at 275 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

## THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-05-01 (knife\_edge)  
Plant limit 18.0 C  
Notice required 3 h  
Forecast skill 0.50 relative to persistence  
Level anchor none  
Condenser bank longest facade  
Switch budget 2 mode changes per day  
Minimum dwell 3 h  
Max dew point 15.0 C (Green Grid WP#46 p.6)

## WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 1 mode change. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 11 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)  
Incumbent 9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)  
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

## EVERY HOUR, AND THE REASON FOR IT

| hh | mode       | safe | binding       | bound  | limit | actual | margin |
|----|------------|------|---------------|--------|-------|--------|--------|
| 00 | mechanical | yes  | switch budget | 12.673 | 18.0  | 11.714 | 0.925  |
| 01 | mechanical | yes  | switch budget | 12.672 | 18.0  | 11.700 | 0.838  |
| 02 | mechanical | yes  | switch budget | 12.734 | 18.0  | 11.679 | 0.740  |
| 03 | mechanical | yes  | switch budget | 12.671 | 18.0  | 11.706 | 0.652  |
| 04 | mechanical | yes  | switch budget | 12.947 | 18.0  | 11.784 | 0.863  |
| 05 | mechanical | yes  | switch budget | 12.666 | 18.0  | 11.236 | 0.801  |
| 06 | mechanical | yes  | switch budget | 12.977 | 18.0  | 11.926 | 0.605  |
| 07 | mechanical | yes  | switch budget | 14.053 | 18.0  | 12.226 | 0.882  |
| 08 | mechanical | yes  | switch budget | 14.895 | 18.0  | 12.256 | 1.107  |
| 09 | mechanical | yes  | switch budget | 16.838 | 18.0  | 13.047 | 1.652  |
| 10 | mechanical | yes  | switch budget | 17.680 | 18.0  | 12.894 | 1.747  |
| 11 | mechanical | no   | dry-bulb      | 18.061 | 18.0  | 14.133 | 1.461  |

|    |            |     |          |        |      |        |       |
|----|------------|-----|----------|--------|------|--------|-------|
| 12 | mechanical | no  | dry-bulb | 18.502 | 18.0 | 15.030 | 1.482 |
| 13 | FREE       | yes | -        | 17.951 | 18.0 | 15.256 | 1.446 |
| 14 | FREE       | yes | -        | 17.946 | 18.0 | 15.815 | 1.322 |
| 15 | FREE       | yes | -        | 17.110 | 18.0 | 14.683 | 1.230 |
| 16 | FREE       | yes | -        | 15.921 | 18.0 | 13.995 | 0.957 |
| 17 | FREE       | yes | -        | 15.441 | 18.0 | 13.444 | 1.118 |
| 18 | FREE       | yes | -        | 13.624 | 18.0 | 12.325 | 1.106 |
| 19 | FREE       | yes | -        | 13.238 | 18.0 | 11.935 | 1.327 |
| 20 | FREE       | yes | -        | 12.391 | 18.0 | 10.583 | 1.484 |
| 21 | FREE       | yes | -        | 12.246 | 18.0 | 10.590 | 1.425 |
| 22 | FREE       | yes | -        | 11.880 | 18.0 | 10.568 | 1.024 |
| 23 | FREE       | yes | -        | 11.403 | 18.0 | 10.674 | 0.772 |

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin  
actual = what the intake temperature turned out to be, from the station record  
margin = group-conditional forecast error for that hour of day, plus plume spread

## THE REASONING, HOUR BY HOUR

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### 00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.673 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.672 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.734 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.672 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.947 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.666 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a

thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**06:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.977 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**07:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.053 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**08:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.895 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**09:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.838 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**10:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.680 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**11:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 18.061 C against a 18.0 C limit, so it fails by 0.061 C. It would take a limit 0.061 C higher, or a bound 0.061 C tighter, to change this hour.

**12:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 18.502 C against a 18.0 C limit, so it fails by 0.502 C. It would take a limit 0.502 C higher, or a bound 0.502 C tighter, to change this hour.

**13:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 17.951 C, which is 0.049 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.446 C, and the margin is measured, not chosen: 1.283 C of group-conditional forecast error for this hour of day, plus 0.0107 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**14:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 17.946 C, which is 0.054 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.322 C, and the margin is measured, not chosen: 1.160 C of group-conditional forecast error for this hour of day, plus 0.0095 C for how much the plume could move if the wind

direction is off by the amount FortyGuard's forecast is actually off by.

**15:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 17.111 C, which is 0.889 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.230 C, and the margin is measured, not chosen: 1.069 C of group-conditional forecast error for this hour of day, plus 0.0090 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**16:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 15.921 C, which is 2.079 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.957 C, and the margin is measured, not chosen: 0.793 C of group-conditional forecast error for this hour of day, plus 0.0117 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**17:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 15.441 C, which is 2.559 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.118 C, and the margin is measured, not chosen: 0.955 C of group-conditional forecast error for this hour of day, plus 0.0106 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**18:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 13.624 C, which is 4.376 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.106 C, and the margin is measured, not chosen: 0.943 C of group-conditional forecast error for this hour of day, plus 0.0114 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**19:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 13.238 C, which is 4.762 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.327 C, and the margin is measured, not chosen: 1.163 C of group-conditional forecast error for this hour of day, plus 0.0118 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**20:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 12.391 C, which is 5.609 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.485 C, and the margin is measured, not chosen: 1.323 C of group-conditional forecast error for this hour of day, plus 0.0097 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**21:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 12.246 C, which is 5.754 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.425 C, and the margin is measured, not chosen: 1.262 C of group-conditional forecast error for this hour of day, plus 0.0107 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**22:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 11.880 C, which is 6.120 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.024 C, and the margin is measured, not chosen: 0.860 C of group-conditional forecast error for this hour of day, plus 0.0114 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**23:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 11.403 C, which is 6.597 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.772 C, and the margin is measured, not chosen: 0.608 C of group-conditional forecast error for this hour of day, plus 0.0124 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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